Customer Shopping Behaviour Analysis Report

Executive Summary

The analysis of 3900 customer records reveals clear patterns in shopping behaviour across gender, age, subscription status, discounts, and product categories. Males drive higher revenue, subscribed customers spend more on average, and discounts significantly influence purchase decisions. Middle-aged and senior groups contribute the most revenue. These insights enable targeted marketing and loyalty programs to boost sales and retention.

Business Problem & Objectives

The company wants to understand purchasing patterns across demographics, categories, and channels to improve sales, satisfaction, and loyalty.

Overarching question: “How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

Methodology (Detailed Focus)

We followed a structured three-step approach as per the deliverables:

1. Data Preparation & Modeling (Python – Jupyter Notebook)

- Loaded raw CSV file.

- Checked and handled missing values (37 missing Review Ratings imputed using category-wise median).

- Standardized column names to snake_case and renamed fields.

- Created new features: `age_group` (quantile-based: Young Adult, Adult, Middle-aged, Senior) and `purchase_frequency_days` (converted text frequency to numeric days).

- Verified and removed redundant column (`promo_code_used` was identical to `discount_applied`).

- Loaded the fully cleaned dataset into PostgreSQL table `customer`.

- Data loading and info check

```
import pandas as pd

df= pd.read_csv('customer_shopping_behavior.csv')
```

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

-Missing value imputation

```python
# Imputation with category median
# df['Review Rating'] =df.groupby('Category')['Review Rating'].transform(lambda x:x.fillna(x.median))
df['Review Rating'] = df['Review Rating'].fillna(
    df.groupby('Category')['Review Rating'].transform('median')
)
```

```python
#check for nulls after imputation with category median
df.isnull().sum()
```

```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
```

-Feature engineering (age_group and purchase_frequency_days)

```python
# Add age_group column based on age of customer
lables= ['Young Adult','Adult','Middle-aged','Senior']
df['age_group']=pd.qcut(df['age'],q=4,labels=lables)
df[['age','age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

-Export to PostgreSQL using SQLAlchemy

```
engine = create_engine(f'postgresql://{username}:{password}@{host}:{port}/{database}')
```

```
# Load the dataframe into postgressql
table_name = "customer"
df.to_sql(table_name,engine,if_exists="replace",index=False)
print(f"Data successfully loaded to table '{table_name}' in database '{database}'")
```

```
Data successfully loaded to table 'customer' in database 'customer_behavior'
```

2. Data Analysis (PostgreSQL – pgAdmin)

Ran 10 targeted SQL queries on the `customer` table to extract business insights:

- Revenue by gender and subscription status
- Impact of discounts on spending
- Top products by review rating and discount usage
- Shipping type comparison
- Loyalty segmentation (New / Returning / Loyal based on previous purchases)
- Age group revenue contribution
- Repeat buyer subscription likelihood
- Top 3 products per category

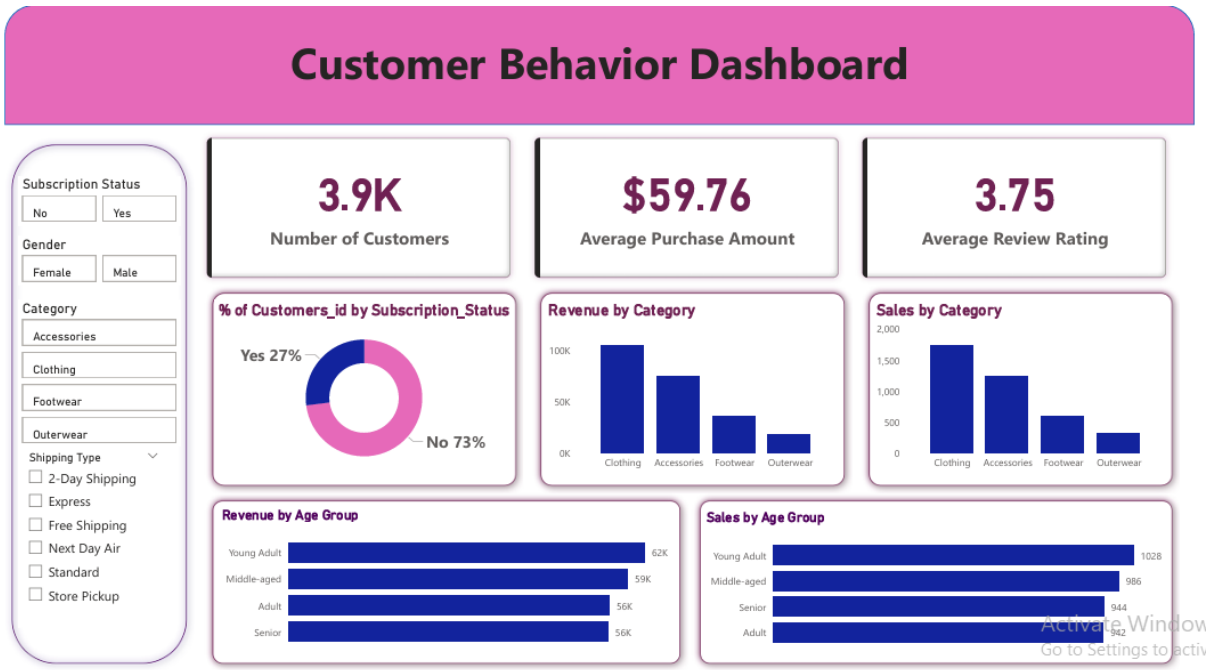
Queries were executed directly on the cleaned data imported as ‘customer’ table to postgres.

3. Visualization & Insights (Power BI Desktop)

Connected to the PostgreSQL `customer` table and built an interactive dashboard with:

- KPI cards for total revenue, average spend, customer count
- Charts showing revenue by gender, age group, category
- Discount vs non-discount analysis
- Top products and shipping comparisons
- Slicers for location, season, and subscription status

- Dashboard screenshot below:



The dashboard enables stakeholders to explore trends dynamically.

Key Findings

- Males generate significantly higher total revenue than females.

- Subscribed customers have higher average spend and total revenue.

- Discounts are used in a high percentage of certain product purchases; some discounted purchases exceed average spend.

- Middle-aged and Senior age groups contribute the highest revenue.

- Repeat buyers (previous purchases >5) are more likely to subscribe.

- Top reviewed products and top volume products per category identified for focus.

Recommendations

1. Prioritize marketing and loyalty programs for male and subscribed customers.

2. Run targeted discounts on high-review items and categories with high discount uptake.

3. Focus product promotions on Middle-aged and Senior segments.

4. Encourage repeat purchases with personalized offers based on previous behaviour.

5. Optimize Express shipping for higher-value orders.

Conclusion

This end-to-end analysis (Python cleaning → SQL insights → Power BI dashboard) provides actionable, data-driven strategies to improve customer engagement and optimize marketing. Implementation can directly impact sales and long-term loyalty.